THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 02

Full Syllabus (Chapters 1–13) | Class 10 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** oxidised	Gains oxygen
2	**(a)** sour	Bases taste bitter
3	**(a)** acidic	CO_2 reacts with $Ca(OH)_2$
4	**(d)** in molten state and aqueous solution	Free ions
5	**(d)** sonority	Sonorous
6	**(a)** Correct	Animal fats contain saturated fatty acids
7	**(b)** C_6H_6	Cyclic
8	**(b)** trachea	Prevent collapse
9	**(b)** growth hormone	Pituitary
10	**(b)** ovary	Fruit
11	**(c)** style	Pistil part
12	**(a)** inherited traits	Heredity
13	**(b)** Student A is correct	Mendel
14	**(c)** +4 D	$1/0.25$
15	**(b)** three	Triangular prism
16	**(c)** opposite to the flow of electrons	Positive to negative
17	**(c)** watt	J/s
18	**(c)** the same at all points	Uniform
19	**(d)** chlorofluorocarbon	CFCs
20	**(d)** it would lead to loss of fertile topsoil	Soil loss

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	It reacts vigorously, releasing heat, forming slaked lime: $CaO + H_2O \longrightarrow Ca(OH)_2$	1+1
22	Bases soluble in water, e.g. NaOH, KOH	1+1
23	False — capillaries are one-cell thick to allow exchange of materials.	2
24	Detects stimuli and starts an impulse; gustatory receptors (tongue)	1+1
25	Gamete: reproductive cell with half chromosomes; zygote: formed by fusion of two gametes, full chromosome number.	2
26	Sperms carrying X chromosome and sperms carrying Y chromosome.	2
27	$m = -2$	2
28	Increase the current; increase the number of turns (use soft iron core).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	(i) $2\,K + 2\,H_2O \longrightarrow 2\,KOH + H_2$ (ii) $ZnCO_3 \longrightarrow ZnO + CO_2$ (iii) $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$	1+1+1
30	Metals form basic oxides and lose electrons; non-metals form acidic/neutral oxides and gain electrons; metals displace hydrogen from dilute acids, non-metals do not.	3
31	Secretes growth hormone and others; controls the secretion of other endocrine glands	2+1
32	Nucleus divides repeatedly within a cyst; each nucleus surrounded by cytoplasm forms many daughter cells released when the cyst bursts	3
33	Gradual weakening of ciliary muscles and loss of lens flexibility with age; corrected by bifocal lenses	3
34	Sprinkle iron filings around a solenoid on cardboard; pattern resembles bar magnet; one end acts as N, other as S (compass test)	3

35	Heat baking soda: $2\,NaHCO_3 \longrightarrow Na_2CO_3 + H_2O + CO_2$; recrystallise: $Na_2CO_3 + 10\,H_2O \longrightarrow Na_2CO_3 \cdot 10\,H_2O$	1½+1½
	OR: Forms salt and water: $Ca(OH)_2 + CO_2 \longrightarrow CaCO_3 + H_2O$ — so non-metal oxides are acidic.	1+1+1
36	Maintains balance; habitat for organisms; regulates climate and water cycle; supplies resources	3
	OR: UV causes skin cancer, eye damage; ozone absorbs most UV radiation in the upper atmosphere	3

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) image at F behind mirror, highly diminished; image between P and F, diminished, virtual (ii) parallel rays diverge after reflection (iii) rear-view mirror	3+1+1
	OR: (i) incident, refracted ray and normal in same plane; $\sin i / \sin r$ = constant (ii) $n = 3 \times 10^8 / 1.5 \times 10^8 = 2$ (iii) refraction at water surface makes the bottom appear raised	2+2+1
38	(i) incomplete combustion deposits carbon (ii) air holes are blocked (iii) butane (iv) produces poisonous CO (v) $2\,C_4H_{10} + 13\,O_2 \longrightarrow 8\,CO_2 + 10\,H_2O$	1+1+1+1+1
	OR: (i) esterification (ii) $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (iii) catalyst/dehydrating agent (iv) saponification — gives ethanol and sodium ethanoate (v) perfumes/flavouring	1+1+1+1+1
39	(i) $P = V^2/R \Rightarrow V^2 = 400 \Rightarrow V = 20$ V (ii) $R = 48.4\ \Omega$; $P = 110^2/48.4 = 250$ W	2+3
	OR: (i) $R \propto l$, $R \propto 1/A \Rightarrow R = \rho l/A$ (ii) $l = RA/\rho = 10 \times 2 \times 10^{-7}/10^{-6} = 2$ m	2+3
40	(i) inhalation: diaphragm flattens, ribs lift, chest cavity enlarges, air rushes in; exhalation: reverse (ii) provides time for O_2 absorption and CO_2 release	3+2
	OR: (i) diffusion of wastes through a semipermeable membrane (ii) nitrogenous wastes, excess salts (iii) both filter blood (iv) no reabsorption in artificial kidney (v) urea	1+1+1+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTER